

An Immersive Analytics Approach to Understanding Human Responses to Room Illuminance

Dongyun Han*
Utah State University

Yimin Zhu†
Louisiana State University

Isaac Cho‡
Utah State University

ABSTRACT

This paper presents an immersive analytics prototype, using the analysis of human responses to room illuminance to address the complex or even 'wicked' nature of indoor environmental condition designs. This prototype replicates real-world environments and allows users to interactively explore and understand the impact of lighting on human behavior. By incorporating a variety of contextual factors such as time of day, external weather conditions, and occupancy, the prototype offers a multifaceted approach to studying human-environment interactions. Users can adjust these factors in real-time, providing a unique and immersive experience that bridges the gap between theoretical models and actual human responses to environmental changes. The ongoing development and refinement of this prototype, guided by user feedback and interdisciplinary collaboration, underscore its potential as a transformative tool in the field of environmental design and human behavior research.

Keywords: Extended reality, virtual reality, occupational support, human factors, contextual factors.

Index Terms: Human-centered computing [Human computer interaction (HCI)]; Interaction paradigms—Mixed/augmented reality; Human-centered computing [Collaborative and social computing]; Collaborative and social computing theory, concept and paradigms—Computer supported cooperative work

1 INTRODUCTION

The design of artificial lighting and use of daylighting significantly impact building energy use [10] and human well-being [8]. Designers consider multiple factors, such as environmental condition requirements, design features, and optimal lighting and balancing energy consumption. These factors create a complex scenario, necessitating the use of analytic models. These models simulate lighting conditions (e.g., [4]) and analyze artificial lighting system usage (e.g., [13]). However, discrepancies often arise between predicted and actual outcomes due to uncertain future occupant behaviors, building operations, and environmental conditions [6]. Understanding these discrepancies is crucial for accurately designing building lighting, fenestration, and HVAC (Heating, Ventilation, and Air Conditioning) systems, as it influences the use of artificial lighting. Therefore, it is essential for designers to comprehend how these factors impact model predictions to make informed design decisions.

Designers face the complex task of predicting how future occupants will interact with building systems, a challenge amplified by the need to meet design objectives like energy efficiency and human well-being. This task is far from straightforward due to the "wicked" nature of many design issues, such as sustainable development [16] and energy efficiency [21]. Accurately defining the optimal type of

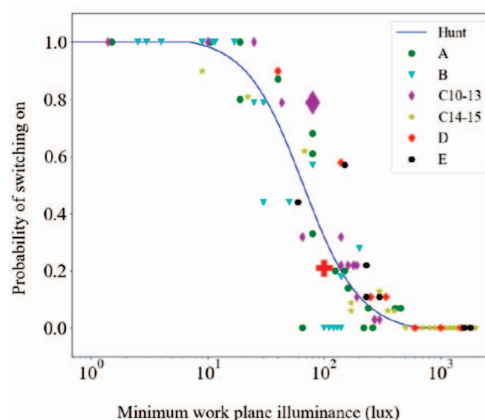


Figure 1: Hunt Model

human-building interactions is difficult, given the uncertainties in design conditions and constraints over a building's lifecycle [5]. These uncertainties, inherent in design processes, make the prediction of human interactions with building systems a particularly challenging aspect of design.

The concept of a wicked problem, complex and ill-structured, is highly relevant in building design, presenting a socio-technical challenge that affects design quality and performance [18]. This issue stems from the gap between the context of model development and application. Rittel [17] highlighted the inadequacy of conventional rational approaches in tackling wicked problems, advocating for an intuitive, second-generation approach. This involves systems thinking and engaging stakeholders through methods like charrettes and collective design. In this approach, experts collaboratively frame the problem, defining specific design conditions and constraints. This framing is crucial as it influences the selection and applicability of models, affecting the prediction of human-building interactions.

Figure 1 illustrates the Hunt model, which predicts the probability of light switch usage at varying illuminance levels [11]. While useful for general analysis, the model requires significant calibration for specific cases. For instance, at 100 lux, the model's prediction may differ by about 100% from actual usage in a specific space type, like C10-13. Such discrepancies might be acceptable in design contexts, where understanding the unique conditions, context, and consequences of design solutions is crucial. Consequently, focusing on the specific design conditions and the relevance of predictive models is a key strategy for enhancing the accuracy of predictions.

2 SENSEMAKING FOR UNDERSTANDING HUMAN BUILDING INTERACTIONS

Sense-making is considered a potential approach to the "wicked" problems of built environment design [22]. As Figure 1 shows, a designer could make better sense of the predictions of a model if the designer is aware of the level of discrepancy in the new design con-

*e-mail: dongyun.han@usu.edu

†e-mail: yimin.zhu@lsu.edu

‡e-mail: isaac.cho@usu.edu

ditions and constraints compared to those of the selected prediction model. Such comparisons involve making sense of the prediction results in the context of new design conditions and constraints. Typically, a model, whether theory-based or data-driven, is developed with a set of parameters describing the conditions and constraints of a mode. For example, studies show that occupant presence [20], light switch location [23], and the type of tasks could have an impact on the use of artificial lighting [3]. Typically, a model's external validity is jeopardized when some of its controlled conditions or constraints, called contextual factors in the following, no longer hold in a new situation.

The research question is about investigating the potential of an immersive environment for exploring various contextual factors, which differ based on specific applications and require domain expertise. These contextual factors are elements that are not a particular model but are conditions or constraints that may still influence the model's predicted results. In the context of lighting, for example, the Hunt model considers work plane illuminance as its primary independent variable. However, other factors like sample size, sample type, task type, and switch location also play a role in light switch usage. These elements are therefore identified as contextual factors in relation to the Hunt model. This list of factors is dynamic and may evolve as further factors are identified and explored.

3 IMMERSIVE ANALYTICS PROTOTYPE FOR HUMAN BEHAVIOR ANALYSIS AND SENSE MAKING USING ROOM ILLUMINANCE

Data visualization, an essential method for interpreting and understanding data, has become increasingly sophisticated with the integration of automated data analytics powered by Machine Learning (ML), Deep Learning (DL), and Artificial Intelligence (AI). This evolution has given rise to visual analytics, a crucial technique for addressing intricate real-world challenges. Visual analytics combines the power of these advanced computational technologies with interactive visual interfaces, thereby enhancing the users' ability to make sense of complex data sets and facilitating more informed and effective decision-making [12, 19]. This synergy of visual representation and automated analysis is pivotal in extracting actionable insights from vast and complex data, making it an indispensable tool in various fields ranging from business intelligence [7] to scientific research [1, 9]. Traditionally, these approaches utilized a desktop environment with a mouse and keyboard for the user interface, thereby offering a limited design space for interaction.

Currently, the field of immersive analytics is rapidly gaining traction, marking a significant intersection between immersive technologies such as Virtual Reality (VR) and Augmented Reality (AR), and the advanced methodologies of visualization and visual analytics [14]. This integration significantly transforms our interaction with data. It introduces novel methods for navigating, comprehending, and extracting insights from intricate datasets. Immersive analytics, as an interdisciplinary domain, seeks to transcend the confines of conventional two-dimensional interfaces. It strives to forge a more dynamic and intuitive milieu for data examination and analysis, thereby enhancing the user's engagement and understanding of complex data structures.

Fig. 3 displays the prototype design for immersive analytics focused on analyzing human behavior in relation to room illuminance. This prototype was developed with Unity version 2021.3.17.f1. The virtual room replicates the same environment used in the study by Chokwithayaa et al. [2], as shown in Fig. 2. Users can enter this virtual office via an Oculus Quest 2 headset. Upon entering the scene, they will find a slider UI on their non-dominant hand and a ray emitting from their dominant hand. This setup allows users to interact with the slider using the ray. Adjusting the slider alters the room's illuminance level, enabling a direct experience of how light influences human behavior in the space.

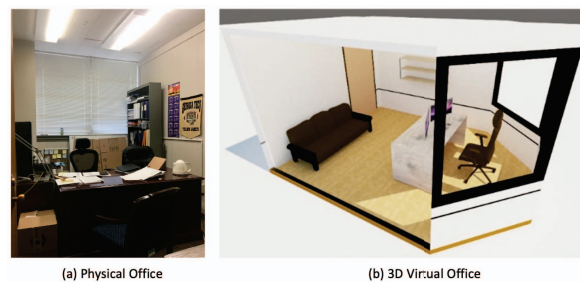


Figure 2: Physical and Virtual Offices.

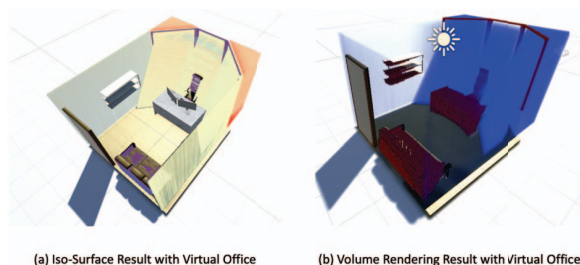


Figure 3: Two 3D visualization examples. (a) and (b) show iso-surface and volume rendering results of light illuminance in the office.

Fig.3-a and b depict an iso-surface and volume rendering results representing illuminance intensities in the room. The iso-surface result shows 3D surface representation of points with equal values in a data distribution. The volume rendering result is projected by using the maximum intensity projection (MIP) technique [15]. MIP is a 3D data projection method that projects the voxels with the highest intensity that fall in the path of parallel rays traced from the viewpoint to the plane of projection in the visualization plane. They are two examples of 3D visualizations for immersive analytics. They would be turned on and off based on user input.

In enhancing the immersive analytics prototype, we will incorporate a range of contextual factors to deepen the analysis of human behavior in relation to room illuminance. Key factors such as external weather conditions, time of day, occupancy levels, and the nature of activities being performed in the space will be seamlessly integrated. This integration will be achieved through advanced data collection methods and interactive visualizations within the virtual environment. Users can interactively modify these factors, such as changing luminance levels or time settings, to observe their impact on lighting preferences and behavior. This enriched virtual environment, facilitated by intuitive user interfaces, allows for a more comprehensive and realistic analysis. Continuous user feedback and iterative design improvements ensure that the prototype remains a dynamic and effective tool for exploring the complex interplay between human behavior and environmental factors in virtual spaces.

4 CONCLUSION

We introduced our work-in-progress immersive analytics prototype, a pioneering tool designed to analyze and make sense of human behavior in relation to room illuminance, aiming to address the complex problem in building design and environmental interaction. By incorporating a range of contextual factors and allowing for real-time manipulation and observation, it offers an unprecedented level of insight into how lighting conditions impact human behavior. The integration of user feedback and continuous iteration ensures that the prototype remains at the forefront of technological and research advancements.

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